

The Probability Gap

Quant Interview Foundations from Zero

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Chapter 1

Counting from Scratch

Everything in probability rests on one act: deciding what “all the ways this can happen” means, and then counting them. Interviewers know this, which is why the first ten minutes of many quant interviews are pure counting — not because the arithmetic is hard, but because sloppy counting is the fastest way to see whether someone actually *thinks* in probabilities or just recites formulas. This chapter builds the counting toolkit from nothing. If some of it feels slow, good — the speed comes later, and it comes from these reflexes being automatic.

1.1 Outcomes, and the multiplication rule

Start with the only object in the chapter that isn’t built from anything else: an **outcome** is one complete way an experiment can go. Roll two dice: the outcome is a pair like $(3, 5)$. Flip three coins: the outcome is a string like HTH . The **sample space** Ω is the set of all outcomes.

The first real question you will ever be asked about a sample space is: how big is it? The central tool is almost embarrassingly simple, and most of this chapter is learning when it applies.

Note 1 (The multiplication rule). If a complete outcome is built in stages, and stage i can go n_i ways *regardless of how the earlier stages went*, then

$$|\Omega| = n_1 \times n_2 \times \cdots \times n_k.$$

Three coins: three stages, two ways each, $2^3 = 8$ outcomes. Three dice: $6^3 = 216$. Ten coin flips: $2^{10} = 1024$. A poker hand dealt as a sequence of five cards: $52 \times 51 \times 50 \times 49 \times 48$ — note the counts shrink, because cards leave the deck as you deal. That shrinking is the signature of sampling *without replacement*, and it matters enough to have its own reflex:

Interview Probe 1 (Spot it instantly). *From a pool of n things taken k at a time, without replacement, in order: the count is*

$$n(n-1)(n-2) \cdots (n-k+1),$$

which is k falling factors. With replacement it would just be n^k . Mixing these two up is the single most common counting error in interviews — state which one you mean out loud before you count.

Where does addition come in? When outcomes arrive as *disjoint cases* rather than stages. If the hand is either a flush or a straight (never both — poker hands are categorized exclusively), the total is a sum. Add for cases, multiply for stages. If the cases can overlap, adding double-counts the overlap, and you get the two-set inclusion–exclusion rule:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Hold on to that minus sign; it grows into the most surprising result in this chapter (Worked Problem 4).

1.2 Order in, order out: permutations and combinations

Suppose the ten candidates for a trading desk are narrowed to three seats. In how many ways can the seats be filled?

Stage by stage: 10 choices, then 9, then 8, so 720 ordered line-ups. But the seats don't come numbered — sitting in them as a group is the same outcome regardless of who filed in first. Every group of three was counted $3! = 6$ times, once per ordering. The honest count is

$$\frac{10 \cdot 9 \cdot 8}{3!} = \binom{10}{3} = 120.$$

This “divide by the number of internal orderings” move is the whole distinction:

Note 2 (Permutations vs. combinations). Choosing k things from n :

$$\underbrace{\binom{n}{k}}_{\text{order does not matter}} = \frac{n!}{k!(n-k)!} \quad \underbrace{\binom{n}{n-k}}_{\text{order matters}} = \frac{n!}{(n-k)!}.$$

The $k!$ in the denominator is doing exactly one job: collapsing the $k!$ orderings of each chosen group into a single outcome.

Interview Probe 2 (“What if order doesn't matter?”). *Any time an interviewer adds that clause mid-question, they are telling you a factor of $k!$ is about to be won or lost. The reflex is not to divide by $k!$ mechanically, but to say why: each unordered group corresponds to exactly $k!$ ordered sequences. If some orderings coincide (repeated elements!), the correction is different — that's when you must slow down and count groupings directly.*

That last warning deserves its own example, because it is a trap that catches people with intact formula recall. How many distinct orderings of the letters in LEVEL? Five letters would suggest $5! = 120$, but the two Es are interchangeable and the two Ls are too. Each distinct word was counted $2! \times 2! = 4$ times, so the true count is $120/4 = 30$. The formula $\binom{n}{k}$ quietly assumes all n objects are distinct; when they are not, count the symmetry by hand.

1.3 Count the complement, count the symmetry

Some events are painful to count directly because “the thing you want” is a scattered union of dozens of cases, while “the thing you don't want” is one clean case. The complement rule fixes this:

$$P(\text{at least one}) = 1 - P(\text{none}).$$

Worked Problem 1 (Two dice, and why 7 is king). *Two fair dice. What is the probability the sum is 7? The direct approach enumerates (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) — six outcomes out of 36, so 1/6. Now look at why those six exist: for every first-die value a , there is exactly one second-die value $b = 7 - a$ that completes the sum. The sum 7 is the only total whose completions never run off the edges of the dice, which is why it beats 6 (which loses (6, 0)-type completions) and 8. **Symmetry as a counting tool:** the table of pairs is symmetric under swapping the two dice, so sums pair up, $s \leftrightarrow 14 - s$, with identical counts — $\Pr(\text{sum} = 6) = \Pr(\text{sum} = 8) = 5/36$. Reading structure like this off the sample space is faster and more convincing in an interview than any enumeration.*

Verify it in code: a three-line simulation of 10^6 dice pairs returns 7 with frequency 0.1667 ± 0.0004 and the 6/8 pair matching to within noise. Every claim in this book gets this treatment — analytic first, simulation second, and the two must agree.

Interview Probe 3 (Push it further). *For three dice, the same edge-effect argument predicts the most likely sums. By symmetry the distribution pairs around 10.5, and the modes are 10 and 11, each achievable in 27 of the 216 outcomes, edging out 9 and 12 at 25 each. Can you say why 10 and 11 beat 9 and 12 without enumerating? (Hint: completions running off the edges again.)*

1.4 Stars and bars: counting distributions

The last core pattern: you have n identical units to split among k recipients. Candy among kids, allocations among traders, photons among states — the objects are identical, so an allocation is described by *how many* each recipient gets, not which ones.

Note 3 (Stars and bars). The number of ways to distribute n identical items among k distinct boxes is

$$\binom{n+k-1}{k-1}.$$

Picture the n items as stars in a row; an allocation is fixed by the $k-1$ dividers (bars) that split the row into k groups. Stars and bars together form a row of $n+k-1$ symbols, and choosing where the $k-1$ bars go fixes everything.

Five candies, three kids: $\binom{7}{2} = 21$ allocations. If each kid must get at least one, hand out the mandatory candy first — n drops to 2, so $\binom{4}{2} = 6$. (Interviewers love that shift: *forced allocations are removed before counting, not after.*)

One honest warning: stars and bars counts allocations of *identical* items. The moment items are distinct — named candies, different-sized orders — you are back in assignment territory (k^n if recipients can repeat, or falling factorials if not). Say “identical or distinct?” out loud before using it.

1.5 Four worked problems

Four problems, each solved two ways: by hand, then confirmed by brute enumeration or simulation in the chapter notebook. This double-check discipline is the point of the course — an interview answer you can also *verify* on the spot is a different tier of confidence.

Worked Problem 2 (The committee). *A quant club has 8 members: 5 traders and 3 researchers. A 4-person committee is chosen uniformly at random. Find the probability it contains (a) any composition, (b) at least one researcher, (c) exactly two researchers.*

(a) Total committees: $\binom{8}{4} = 70$ — this is the denominator for every part.

(b) Direct count of “at least one researcher” is three cases (one, two, or three researchers). Complement is one case: a committee with no researchers is all-trader, $\binom{5}{4} = 5$. So

$$P(\text{at least one researcher}) = 1 - \frac{5}{70} = \frac{65}{70} \approx 0.929.$$

(c) Exactly two researchers: choose which 2 of the 3 researchers ($\binom{3}{2} = 3$) and which 2 of the 5 traders ($\binom{5}{2} = 10$). Cases multiply because the choices are independent stages:

$$P(\text{exactly 2 researchers}) = \frac{3 \times 10}{70} = \frac{30}{70} \approx 0.429.$$

Verify: the notebook enumerates all $\binom{8}{4} = 70$ committees with `itertools.combinations` and counts each event directly — 65 and 30, matching exactly. No simulation tolerance needed; with small spaces you can hold the entire sample space in memory, and you should.

Worked Problem 3 (The birthday problem, from counting). *How many people must be in a room before a shared birthday is more likely than not? (Assume 365 equally likely days, no leap days, independent people.)*

Count the complement: no shared birthdays. Person 1 can have any day. Person 2 must avoid it (364 safe days of 365), person 3 must avoid two days, and person k must avoid $k-1$ occupied days. The stage-by-stage probability of no collision among n people is

$$P(\text{no shared}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365-(n-1)}{365} = \prod_{j=1}^{n-1} \left(1 - \frac{j}{365}\right).$$

So $P(\text{shared}) = 1 - \prod_{j=1}^{n-1} (1 - j/365)$. Running the product: at $n = 23$ it first crosses one half,

$$P(\text{shared among } 23) \approx 0.5073.$$

Twenty-three — far smaller than most people’s gut says, and the gap between gut and math is exactly why interviewers ask it. The product falls faster than intuition expects because pairwise collision chances grow quadratically: n people means $\binom{n}{2}$ pairs, each a small collision chance, and $\binom{23}{2} = 253$ pairs is a lot of shots at a hit.

Verify: the notebook simulates 100,000 rooms of 23 random birthdays: the shared-birthday frequency lands within ± 0.01 of the exact 0.5073 (a tolerance set honestly by the simulation’s own $\frac{1}{\sqrt{N}}$ error scale — Chapter 7 makes that calculation exact, and the test suite encodes it so no test ever passes by luck).

Worked Problem 4 (The hat-check problem, counting version). A restaurant loses the tickets for $n = 5$ hats and returns them uniformly at random — one hat per guest. How many returns leave nobody with their own hat?

Total returns: $5! = 120$. Counting “nobody gets theirs” directly is a mess of cases; instead, build up with inclusion–exclusion. Let A_i be the event guest i gets their own hat. We want the total minus the union $|A_1 \cup \dots \cup A_5|$. Summing singles counts $|A_i| = 4!$ each; pairs $|A_i \cap A_j| = 3!$; and so on, alternating signs:

$$D_5 = 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right) = 120 \cdot \frac{44}{120} = 44.$$

Forty-four of the 120 returns leave everyone hatless-but-happy, so $P(\text{nobody right}) = 44/120 \approx 0.367$.

The alternating sum is not a trick; it is the two-set rule $|A \cup B| = |A| + |B| - |A \cap B|$ extended to five sets, with signs alternating because each overlap layer corrects the over- or under-count of the layer above. And the punchline hiding in the formula: as $n \rightarrow \infty$ the bracket tends to $e^{-1} \approx 0.368$. The chance that nobody is right barely moves — $D_4 = 9$ gives $9/24 = 0.375$, already close. Expectation is Chapter 3’s business, where this problem returns with a one-line answer ($E[\text{correct hats}] = 1$, always) via the indicator trick; remember this counting route, because the two methods must and do agree.

Verify: the test suite checks $D_5 = 44$ against brute-force enumeration of all 120 permutations, plus the recursion $D_n = (n-1)(D_{n-1} + D_{n-2})$ — two independent roads to the same number.

Worked Problem 5 (Straight flushes, counted honestly). Because interviewers love card counting and it exercises every tool at once: how many 5-card hands are straight flushes? (Ace can be low, as in A-2-3-4-5; no wildcards. Royal flushes included.)

Work in stages. A straight flush is fixed by two choices: its rank sequence and its suit. Rank sequences: the lowest card of a straight runs from A(low) up to 10 (for 10-J-Q-K-A), so there are 10 sequences. Suits: 4. Total hands: $10 \times 4 = 40$ out of $\binom{52}{5} = 2,598,960$, so

$$P(\text{straight flush}) = \frac{40}{2,598,960} \approx 1.54 \times 10^{-4} \quad (\text{about 1 in 65,000}).$$

The structure to internalize: decompose into independent features (sequence, suit), count each, multiply. The trap is double counting — e.g., counting “royal flush separately” would count those 4 hands twice. When your counting scheme has a name for a special case, check whether the special case is already inside the general count before adding it.

Verify: the notebook enumerates all 2,598,960 hands once with *itertools* (a few seconds of compute) and the straight-flush count is exactly 40 — along with every other poker hand class, which becomes the test suite’s reference table.

1.6 Where this is going

You now have every counting pattern the rest of the course stands on: multiply across independent stages, add across disjoint cases, divide out internal order, flip to the complement when “at least one” hurts, exploit symmetry before enumerating, and stars-and-bars when items are identical. Chapter 2 turns counts into probabilities conditioned on partial information — the single biggest jump in interview difficulty — and Chapter 3 retools everything here into the expectation workhorse that half of all quant interview questions run on.

Drills for this chapter. (1) Three dice: enumerate the counts for sums 9–12 and confirm the 27/25 claim of Section 1.3. (2) How many ways to deal 4 cards each to 3 players from a 52-card deck? ($52!/(40!(4!)^3)$ if the players are named — say which assumption you used.) (3) Redo Worked Problem 2 for “at least two researchers” ($35/70$ — complement again). (4) Distinct rearrangements of BANANA? ($60 = 6!/(3!2!1!)$.) Check every answer against the chapter notebook before moving on; that habit — *never trust an unverified count* — is the course in one sentence.